

Shubhan Chari

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Machine Learning/software student with hands-on experience in training and deploying ML solutions, building ETL, and Industrial Data Analytics. Expert in Python & SQL with a focus on building data solutions for business, tech, and engineering.

EDUCATION

University of Illinois at Urbana-Champaign - GPA: 4.0 / 4.0

Expected May 2028

Bachelor of Science in Statistics & Computer Science | Minors in Data Science and Econometrics

Relevant Coursework: Data Structures, Modeling in Data Science, Intro to Databases (SQL), Reading & Writing Data

SKILLS

- Technical: Python, SQL, C++, SAP, Git, Excel w/ Macros, Power BI, PLC Ladder Logic, Office Suite, Seeq
- Tools/Libraries: PyTorch, Scikit-learn, Pandas, Numpy, Matplotlib, MongoDB, PostgreSQL, Docker
- Soft Skills: Data Storytelling, Collaboration, Analytical Decision Making, Fast Learner, Teamwork

WORK EXPERIENCE

International Fragrances & Flavors (Protein Products Manufacturing Division)

Gibson City, IL

Engineering Co-Op - Computer Science

June 2025 - Jan 2026

- Architected a **Python-based ETL pipeline** to automate data analysis & engineering workflows, extracting signal data from **SQL-based** MES databases and transforming raw **XML** into **DataFrame** at **2x efficiency**
- Designed **low-latency process dashboards** with batch processing using **Seeq Asset Trees** and **Python** for scalable, real-time process performance monitoring and analysis for **200+** process variables and equipment
- Applied **signal processing and statistical analysis** (linear regression, anomaly detection) to diagnose and document industrial process failures, report the root causes, and develop solutions to prevent future issues
- Built product recycle input-output data model to optimize screw conveyor output, saved **\$1000/day** in feed
- Developed **PLC process control** software and interlock systems to optimize and reduce error in product recycle input-to-output and aspiration systems, decreasing equipment and operational error by **over 80%**

TECHNICAL PROJECTS

Cooking Chatbot - Machine Learning Engineer - <https://cookingchatbot.vercel.app/>

Jan 2026 - present

- Fine-tuned OpenAI's **CLIP** vision transformer with **PyTorch** and WebDataset sharding, utilizing a prompt ensemble pipeline to optimize zero-shot classification accuracy, raising top-5 accuracy from **45% to 90%**
- Developed robust data cleaning pipelines to normalize 200,000 raw ingredients into 597 canonical classes with substring-based grouping and logic parameters to standardize and filter a 2 million-row recipe dataset
- Implemented backend database systems on **PostgreSQL** (Aiven) to manage **1.5M+** recipes, ingredient lookups, and user information and preferences integrated with Google OAuth for secure authentication

Fake News Detection System - <https://fake-news-detector-sc.vercel.app/>

Nov 2025 - present

- Deployed serverless misinformation detector (FastAPI/Vercel): paired SGD-SVM with TF-IDF n-grams for articles (**~97% acc.**) and XGBoost with 768-dim MPNet sentence embeddings for tweets (**0.89 ROC-AUC**)
- Diagnosed statement-level data leakage inflating XGBoost ROC-AUC to 0.99; redesigned evaluation using GroupShuffleSplit CV across 1,058 unique statements, validating a leak-free 0.89 AUC on 130K+ tweets
- Engineered real-time routing pipeline (FastAPI): scrapes raw URLs via BeautifulSoup & Twitter APIs, generating 46 lexical/NER features and fetching Hugging Face MPNet embeddings for low-latency inference

UIUC Study Spots (Data Science Development Group) - <https://uiuc-study-spots.vercel.app/> *Feb 2026 - May 2026*

- Architected a demand forecasting system for 8 campus facilities, utilizing **Gaussian curve modeling** to simulate student occupancy, integrating building hours to create hourly predictions for a 16-week semester
- Transformed unstructured "Google Popular Times" data into a baseline dataset for demand curve modeling
- Developed exam pressure multiplier & spillover redistribution, with **70%** attrition and **30%** reallocation

ACTIVITIES

Marketing Chair - Association for Computing Machinery at UIUC: Produce and manage social media strategy and content for 2,000+ followers, reaching 80,000+ views/month to drive ACM event participation and retention

Treasurer - Open Source at Illinois: Managed organizational budgeting & spending for tech talks, workshops and open-source projects; Increased student engagement in events and projects by **over 50%** with targeted outreach